Human-Robot Collaborative Manufacturing

Behzad Sadrfaridpour (PhD)

Adviser: Dr. Yue Wang

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Overview

- The recent emergence of safe, lightweight, human-friendly, and flexible robots has opened a new realm for human-robot collaboration (HRC) in manufacturing.
- These developments improve human-robot interaction (HRI) to the extent that robots are perceived as social beings with which humans interact rather than simple tools.
- For such robots with the new human-robot interaction (HRI) functionalities to interact closely and effectively with a human coworker, new HRI-based control criteria are demanded.
- The aim of this research is to integrate HRI factors (both physical and social interactions) into the robot motion controller for human-robot collaborative assembly tasks in a manufacturing hybrid cell.

Motivation/State of the Art

Typical assembly tasks: pick and place task and handover task

Strengths

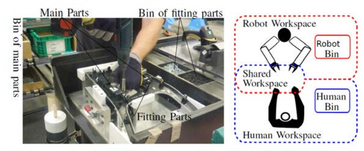
- Assist in repetitive tasks
- Can co-manipulate heavy objects
- New flexible robots
  - Do not need programming
  - Can learn by demonstration

Weaknesses (problems)

- Human performance and speed changes
- How fast the robot should accomplish the task based on HRI?
- Human's comfort and acceptance?
- Safe and efficient handover

Proposed solutions

- Design a framework to control the robot speed along a given path based on HRI considering
  - Human input (manual control)
  - Physical interaction (pHRI)
  - Social interaction (sHRI)
- Develop a safe and adaptive robot-to-human hand over strategy
- Develop a framework for human-robot co-manipulation



(a) A conventional manufacturing cell (Photo courtesy of BMW Spartanburg, SC). (b) An HRC hybrid manufacturing cell.

Progress Made

Trust Model

- The model for the human's trust to robot in manufacturing settings was found.
- The human trust depends on previous trust, robot and human performance and fault.
- The preliminary results were presented in 2014 AAAI Spring Symposium Series.
- The final results of human-in-the-loop experiments for trust model were published in a Springer book chapter.

Manipulation Framework

- An HRI-based framework were developed for controlling the speed of a collaborative robot in a hybrid manufacturing cell.
- The preliminary results were presented and won the best student conference paper award in 2016 IEEE Conference on Automation Science and Engineering (CASE).
- Facial expressions and trust-based path planning were added to the framework.
- The final results of the thorough statistical analysis of the implementation of the framework were published to IEEE Transaction of Automation Science and Engineering (T-ASE).

Handover Strategy

- A trust-based hand-over strategy was developed to optimize the safety and efficiency of the interaction during human-robot handovers.
- The robot uses its measurements of the human performance and fault during the interaction to evaluates artificial trust of robot to human.
- The final results of this study were submitted to IEEE Transaction of Human-Machine Systems (THMS) and is currently under review.

Co-Manipulation Framework

- A mobile base robot was integrated to the collaborative robot for performing the experiments with a mobile manipulator.

Intellectual Merit and Broader Impacts

- This study proposes novel trust-based control architectures and frameworks for human-aware, smooth and effective HRI in manufacturing environments.
- The result of this study contributes to the transition of the current conventional robots into the more capable collaborative robots.

Acknowledgements

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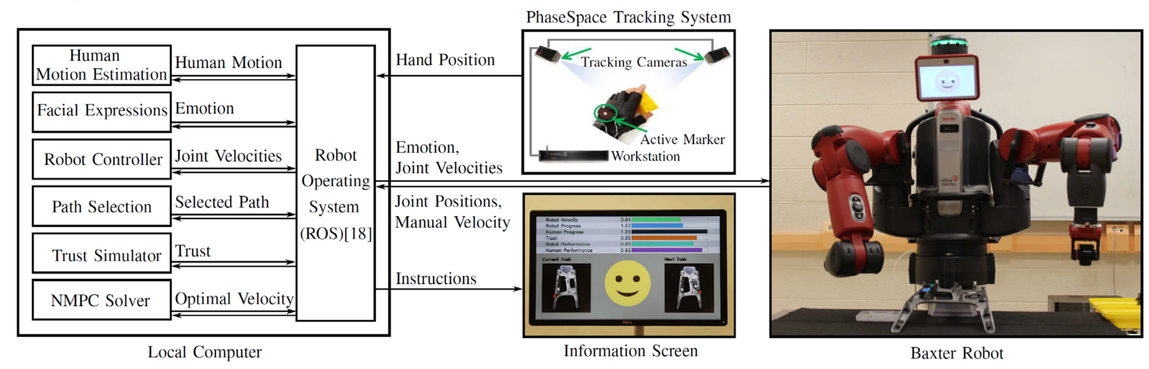
Research Plan

- ✓ Dynamics of human's trust in robot during the interaction is modelled first. This model is evaluated and verified through a series of IRB-approved human-in-the-loop experiments.
- ✓ An HRI-based framework is developed for controlling the speed of a collaborative robot performing pick and place tasks alongside a human in a hybrid manufacturing cell.
- ✓ A framework that considers both pHRI and sHRI is developed for planning and controlling the motion of the robot performing hand-over tasks to the human in a hybrid manufacturing cell.
- ✓ A series of IRB-approved human-in-the-loop experimental studies is conducted to evaluate the impact of implementation of the frameworks on overall efficiency and HRI criteria such human workload and trust and robot usability.
- ✓ A Trust-based hand over strategy is developed and evaluated similarly to previous steps.
- ➢ A testbed is developed for co-manipulation of a human and mobile robot
- ➢ Impact of level of autonomy on the interaction is studied by conducting a preliminary human-in-the-loop experiments.
- ➢ A shared-control strategy is proposed for performing human-robot co-manipulation tasks.

Illustrative Figures

Manipulation Framework



The architecture of the integrated framework and illustration of the experiment setup.

Step 1: Choose a path based on both HRI and efficiency

Step2: Controlling the robot with a given velocity

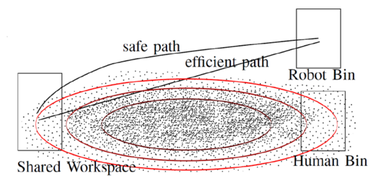
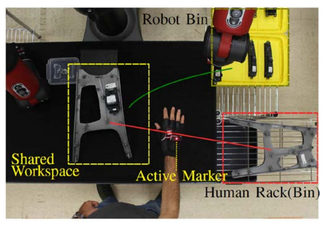
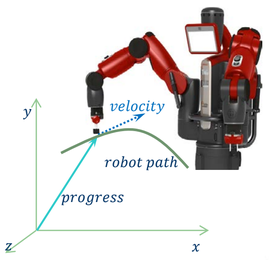


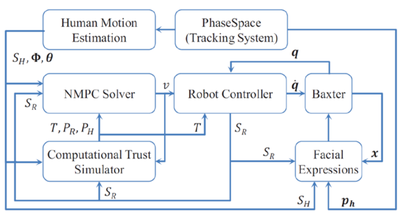
Illustration of sample data of human movements in the workspace: an efficient robot path versus a safe robot path.



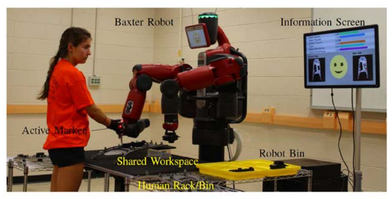
Robot (green) and human (red) sample paths.



Step 2: Adjusting the robot velocity based on Human-Robot Collaboration System



Block diagram of the HRC control framework.

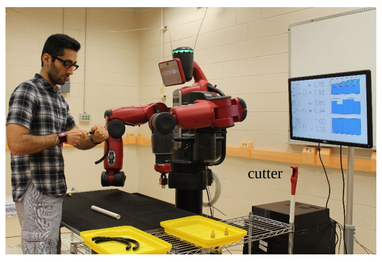


The experiment scenario.

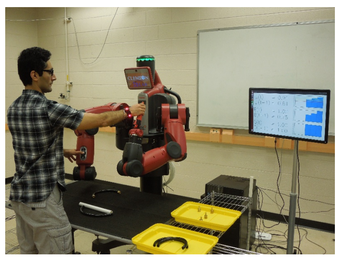
(a) Happy Face
(b) Worried Face
(c) Bored Face

Robot emotions

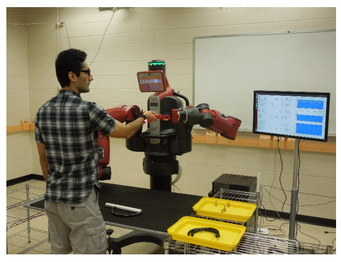
Handover Strategy



Robot manipulates the hoses and parts to the human and hands over a cutter to the human if necessary.



Cautious handover.



Efficient handover.

Conclusions

- To this end, human trust's in robot was modelled.
- Moreover, a trust-based framework was developed for controlling the speed of the robot.
- Furthermore, a trust-based strategy was developed for hand over tasks.
- Finally, a framework for smooth and efficient co-manipulation tasks will be developed to excel the human-robot interaction in collaborative manufacturing.

Interdisciplinary & Intelligence Research

For More Information Contact
Behzad Sadrfaridpour at bsadrfa@g.clemson.edu

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